

PROCESS DESIGN PRINCIPLES I

CHE F314 · BITS Pilani · I-Semester 2026-27

Short Notes & Formula Sheet

Process synthesis · The hierarchy of decisions

Case studies: HDA, vinyl chloride, acetone recovery

Absorber design · Economic optimisation · Energy integration

Lecture-1, 04-08-2026

Compiled from: the five course slide decks in Slides/ — L-1.pdf (22 pp), L-2.pdf (35 pp), L-4.pdf (42 pp), EDM.pdf (97 pp) and EIA_Selected_Slides.pdf (6 pp).

Companion documents: PDP-Problems-and-Solutions.pdf works every numerical the decks pose (P1–P7); pdp-answers-check.py reproduces every number quoted below and exits 0.

One caveat about a source: EIA_Selected_Slides.pdf is the only deck in the set with *no text layer* — its six pages are images, so section 5 carries only what is legible there and is correspondingly shorter than the others. The other four decks were read directly from their text.

What is inside

Topic	Deck
What process design is; synthesis vs analysis; why design problems are under-defined; the five-level hierarchy of decisions; batch vs continuous	L-1
The hierarchical approach; the five synthesis steps; the HDA case study; ordering rules for VRS, LSS and EIA; input–output economics	L-2
Pathway economics and gross profit; the vinyl chloride case study; pressure decisions; block, process and instrumentation flow diagrams	L-4
Economic potential; recovery-route selection; the acetone case study; absorber heuristics and the Kremser equation; optimum solute loss; approach temperature; minimum trays	EDM
Composite curves, the pinch, exchanger area and the grid diagram	EIA

Notation used throughout

L, G	solvent and gas molar flows	m	equilibrium slope, $y^* = mx$
N	theoretical stages	x, y	liquid, gas solute mole fractions
γ_s	solvent activity coefficient	P_s°	solute vapour pressure
EP	economic potential	TAC	total annualised cost
U	overall heat-transfer coefficient	A	exchanger area
ΔT_{lm}	log-mean temperature difference	ΔT_{min}	minimum approach temperature
Q	exchanger duty	α	relative volatility
N_{min}	minimum number of trays	λ	latent heat of vaporisation
F, c_p	stream mass flow, heat capacity	C_A, C_N, C_s	area, stage and solute costs

1 Introduction and the Hierarchy of Decisions

Source: L-1.pdf.

1.1 What process design is

- **Chemical process design is about finding a sustainable process that can convert the raw materials to the desired chemical products.** The deck's own framing: *“a combination of science and art in a creative activity.”*
- **The purpose of engineering is to create new material wealth**, and in chemical engineering that is accomplished *via the chemical transformation and/or separation of materials*. Every flowsheet decision in this course is one of those two.
- **Scale of the opportunity:** about **50 % of the chemical products sold were developed during the last few decades** (30–40 years) — the deck's evidence that the engineering effort works.
- **Process and plant design** is the creative activity of generating ideas and translating them into equations and processes, either to produce new materials or to *significantly upgrade the value of existing ones*.

1.2 Synthesis versus analysis

- **Synthesis** — *building* the flowsheet. **Analysis** — *understanding* a given one. The two need different tools, and the course is mostly synthesis.
- The goal of a conceptual design is twofold: to find the **best process flowsheet**, and to estimate the **optimum design conditions**.
- **Why synthesis is hard:** design problems are *under-defined*, and **10^4 – 10^9 alternatives can be generated for a single product plant**. With that many possibilities and only a small chance of success, a strategy is not optional.
- **The missing information must be assumed:** what type of process units to use, how they are interconnected, and what temperatures, pressures and flow rates are required. That act of assuming *is* the synthesis activity.

heuristics	→ set structure and some design variables	first resort
shortcut methods	→ where heuristics run out	second resort

1.3 The objective

- Out of the 10^4 – 10^9 alternatives we must find the one that has the **lowest cost**, is **safe**, **satisfies environmental constraints**, and is **easy to start up and operate**. Those four are the standing test applied to every case study in this course.

1.4 The hierarchy of decisions

Five levels, taken in order. Each level's answer constrains the next, which is why they cannot be taken out of sequence.

1. **Level 1** — **Batch vs continuous**.
 2. **Level 2** — **Input–output structure** of the flowsheet.
 3. **Level 3** — **Recycle structure** of the flowsheet.
 4. **Level 4** — **General structure of the separation system**, split into **4a. VRS** (vapour recovery system) and **4b. LSS** (liquid separation system).
 5. **Level 5** — **Energy integration analysis (EIA)**.
- **Level 1, batch vs continuous:** batch is preferred for *small capacities*, *campaign operation*, *fouling* services, or where there is a *proven batch market*; continuous otherwise.

2 The Hierarchical Approach — the HDA Case Study

Source: L-2.pdf.

2.1 The five synthesis steps

Synthesis as the elimination of differences, in order:

1. **Eliminate differences in molecular type** — by chemical reaction.
2. **Distribute the chemicals** — match sources to sinks.
3. **Eliminate differences in composition** — by separation.
4. **Eliminate differences in temperature, pressure and phase**.
5. **Integrate the tasks** into unit operations.

2.2 The HDA process

- **Reaction:** toluene + $H_2 \rightarrow$ benzene + CH_4 .
- **Reactor conditions, and why:**

1150–1300 °F	below the range the reaction is too slow; above it hydrocracking takes over
500 psia ($\approx$ 34 atm)	operating pressure
excess H_2	suppresses coking
purge	required because CH_4 builds up in the recycle loop
toluene recycled	unconverted feed returns to the reactor

- **Boiling points set the separation sequence** (all in °F):

	H ₂	CH ₄	benzene	toluene	diphenyl
T_b (°F)	−423	−259	176	232	491

The gap between −259 and 176 °F is enormous, so a **flash / light-gas split comes first** and **distillation follows**. Reading a boiling-point table before choosing separators is the habit this deck is teaching.

2.3 Ordering rules that the hierarchy forces

- **The VRS is designed *before* the LSS**, because the VRS's losses fix the purge composition — and the purge composition is an input to the liquid separation.
- **EIA is done *last***, after the distillation train is fixed, because heat-exchanger network synthesis needs *every* stream's flow, composition and temperature. Those do not exist until the earlier levels are settled.

2.4 Why input–output structure dominates the economics

- **Raw materials are normally 33–85 % of total product cost.** Nothing else in the flowsheet comes close, which is why Level 2 is decided before anything is sized.
- **Overall material balances are therefore evaluated first**, and any alternative whose *products are worth less than its feeds* is discarded immediately — no amount of downstream cleverness rescues it.

3 Pathway Economics — the Vinyl Chloride Case Study

Source: L-4.pdf.

3.1 Gross profit as the pathway selector

$$\text{gross profit} = \text{value of products} - \text{cost of reactants} \quad \text{per lb of vinyl chloride}$$

- Paths are compared on **lb per lb of VC from stoichiometry, times the price in cents per lb**. The deck's prices:

	ethylene	acetylene	Cl ₂	VC	HCl	water
cents/lb	18	50	11	22	18	0

3.2 Path-3 flowsheet, balanced for 100 000 lb/h VC

Stream	lb/h
vinyl chloride product	100 000
Cl ₂ feed	113 400
C ₂ H ₄ feed	44 900
DCE to pyrolysis	158 300
DCE recycle (from 60 % per-pass conversion)	105 500

Chlorination runs at **100 % conversion**; pyrolysis at **60 % per pass**, which is what creates the DCE recycle. Note that the recycle is larger than the fresh feed of either reactant — a typical and easily missed consequence of a modest per-pass conversion.

3.3 Pressure decisions, and the reasoning behind each

Unit	Pressure	Why
chlorination	1.5 atm	above atmospheric so there is no air ingress
pyrolysis	26 atm	—
first column	12 atm	HCl boils at −26.2 °C; bottoms bubble point 93 °C — keeps operation away from the critical region
second column	4.8 atm	distillate at 33 °C condenses on cooling water ; bottoms at 146 °C on medium-pressure steam

- **Feed temperature ≥ 35 °C.**
- **Pump liquids, not vapours** — molar volume is far smaller, so the compression work is far lower.
- **Cool the pyrolysis effluent by partial condensation** to set the feed condition for the column.

Every pressure in this table is chosen to make a *utility* available at ordinary conditions: cooling water on one condenser, medium-pressure steam on one reboiler. That is the whole point of choosing column pressure early.

3.4 The three levels of flow diagram

BFD	block flow diagram — functional blocks only
PFD	process flow diagram — all major units, numbered stream diamonds, stream table, utilities, equipment summary
P&ID	pipng and instrumentation diagram — <i>every</i> instrument and control loop; the construction document

4 Economic Decision Making — the Acetone Case Study

Source: EDM.pdf (97 slides).

4.1 Economic potential

$$EP = \text{product value} - \text{raw-material cost} \quad \text{the first screen on any project}$$

- **The case:** 10.3 mol/h of acetone in 687 mol/h of air. This gives

$$EP = \text{Rs } 5.26 \text{ Cr/yr } (\approx \$1.315 \times 10^6/\text{yr at } 8150 \text{ h/yr})$$

- 8150 **h/yr** is the annual operating hours used throughout this deck; it converts any hourly figure to an annual one.

4.2 Choosing the recovery route

- **The alternatives:** condensation (high pressure / low temperature), absorption, adsorption, membrane, and reaction.
- **The heuristic:** if the solute is $< 5 \text{ mol } \%$ in the gas, **adsorption is cheapest**.

$$\frac{10.3}{697.3} = 1.47 \% \implies \text{the heuristic applies here}$$

4.3 The four absorber heuristics

- H1** If a raw-material component can act as the absorber *solvent* (here, water), consider **feeding the process through the gas absorber**. This justifies discarding part of the process water: **\$25 600/yr** against the economic potential above.
- H2** Recover $> 99 \%$ of valuable material; use 99.5% as the first guess.
- H3** For an **isothermal dilute** absorber, take $L = 1.4 mG$.
- H4** **Avoid adiabatic absorbers** — their minimum L can be $\sim 10\times$ larger, $L \approx 14 mG$.

H3 and H4 are the same instruction from two directions: keep the absorber isothermal, because letting it heat up costs an order of magnitude in solvent.

4.4 Equilibrium slope and solvent rate

$$m = \frac{\gamma_s P_s^\circ}{P} \quad \text{equilibrium slope from the solvent activity coefficient}$$

$$m = \frac{6.7 \times 229}{760} = 2.02 \quad \text{acetone–water at } 25^\circ\text{C, } 1 \text{ atm}$$

$$L = 1.4 mG = 1943 \text{ mol/h} \quad \text{from H3}$$

4.5 The Kremser equation

Valid for an *isothermal, dilute* absorber with straight equilibrium and operating lines — exactly the conditions H3 and H4 arrange.

$$N + 1 = \frac{\ln \left[\left(\frac{y_{in} - mx_{in}}{y_{out} - mx_{in}} \right) \left(1 - \frac{mG}{L} \right) + \frac{mG}{L} \right]}{\ln(L/mG)}$$

- With $L/mG = 1.4$ and $x_{in} = 0$ (clean solvent):

$$99\% \text{ recovery} \implies N \approx 10 \text{ (exact 10.1);} \quad 99.9\% \implies N \approx 16$$

- **Keep $1 < L/mG < 2$.** The reasoning is the part worth remembering: $L/mG < 1$ needs *infinitely many trays*, while $L/mG > 2$ *starves the downstream distillation column's feed*. 1.4 is the trade-off, not an arbitrary factor.

4.6 Optimum solute loss from the TAC model

$$\text{TAC} = C_s G y_{in} \left(\frac{y_{out}}{y_{in}} \right) 8150 + C_N [N + 2] \quad \text{solute loss + stages}$$

- Minimising gives a result of the form $\frac{y_{out}}{y_{in}} \approx \sqrt{\frac{C_N \ln(L/mG)}{8150 C_s G y_{in}}}$, and numerically

$$\boxed{\frac{y_{out}}{y_{in}} \approx 0.004} \quad \text{i.e. 99.6\% recovery}$$

- **This optimum is insensitive to $\pm 100\%$ changes in the cost parameters.** That insensitivity is the practical payoff: H2's "recover $> 99\%$ " is not a guess, it is within a factor of two of the true optimum, so the heuristic is safe to apply without a costing study.

4.7 How the design variables move the answer

Change	Effect
double P_T	halves L (smaller still), same number of trays
hotter solvent	raises γ_s and P_s° , so raises m and therefore L
use MIBK ($\gamma = 1$)	shrinks L , but the solvent's own cost kills the saving

All three say the same thing: m is the lever. Anything that lowers m lowers L , and pressure and temperature both act on m through P_s° .

4.8 The approach-temperature heuristic

- **Rule of thumb:** a 10°F approach, applied from ambient up to the boiling point of organics.
- **The deck then tests its own rule** on a waste stream with $F = 51000 \text{ lb/h}$, $c_p = 1.0$, $U_c = 30$ and $U_s = 20 \text{ Btu/h}\cdot\text{ft}^2 \cdot ^\circ\text{F}$, and $C_A = \$11.38/\text{yr}\cdot\text{ft}^2$. The optimum comes out near 21.5°F — so the 10°F rule is *conservative*. Section 6.1 quantifies what that conservatism costs.

4.9 Minimum trays: back-of-envelope versus Fenske

$$N_{min} \approx \frac{T_{b1} + T_{b2}}{3(T_{b2} - T_{b1})} \quad \text{back-of-envelope, boiling points only}$$

$$N_{min} = \frac{\log \left[\frac{x_D}{1 - x_D} \cdot \frac{1 - x_B}{x_B} \right]}{\log \alpha} \quad \text{Fenske}$$

- The deck's claim is that the first expression *is* Fenske's equation when $\lambda/R \approx 5080 \text{ K}$. Section 6.2 tests that claim numerically rather than accepting it.

5 Energy Integration

Source: EIA_Selected_Slides.pdf — six image-only pages, so this section is shorter than the others and carries only what is legible there.

5.1 Composite curves and the pinch

- **Hot and cold streams are drawn as enthalpy–temperature composite curves.** The closest vertical distance between them is ΔT_{min} ; the deck works with the 10°F approach of §4.8.
- **The pinch** is where the two curves touch at ΔT_{min} . It splits the problem in two:

above the pinch: only hot utility

no heat crosses the pinch

below the pinch: only cold utility

no heat crosses the pinch

5.2 Exchanger duty, area and the grid diagram

$$Q = U A \Delta T_{lm}$$

$$\implies A = \frac{Q}{U \Delta T_{lm}}$$

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}$$

log-mean temperature difference

- **The grid diagram** records the matches: exchangers appear as circles or lozenges between stream lines, with heaters and coolers at the ends. **Matches that cross the pinch are the retrofit targets** — each one is destroying available driving force.

6 Worked from the Decks

Both examples below are reproduced in full by `pdp-answers-check.py`, which exits 0.

6.1 P1 — How much does the 10 °F rule actually cost?

EDM.pdf, slides 94–95. Waste stream $FC_p = 51\,000$ Btu/h·°F; heat available above 100 °F is 1.3566×10^7 Btu/h, enough to raise 14 529.3 lb/h of steam.

T_1 (°F)	dt_s	dt_c	A_s/ft^2	A_c/ft^2	TAC (\$/yr)
277	10.0	157.0	5846	5637	49 745
280	13.0	160.0	5177	5656	46 207
285	18.0	165.0	4347	5688	43 552
290	23.0	170.0	3722	5720	43 220
295	28.0	175.0	3220	5750	44 286
300	33.0	180.0	2801	5781	46 286
310	43.0	190.0	2126	5840	52 124
320	53.0	200.0	1593	5897	59 552
340	73.0	220.0	777	6005	77 190
360	93.0	240.0	159	6107	97 017

Refined by optimisation rather than read off the table:

$$T_1^{\text{opt}} = 288.5 \text{ °F}$$

approach, steam-generator cold end = 21.5 °F

$$A_s = 3894.02 \text{ ft}^2$$

$$A_c = 5710.18 \text{ ft}^2$$

area cost = \$109 296

CW cost = \$23 675

steam credit = \$89 828

$$\boxed{\text{TAC}_{\text{opt}} = \$43\,143/\text{yr}}$$

TAC at the 10 °F rule = \$49 745/yr

penalty against the optimum = 15.3%

- **The rule is conservative but not free.** Designing to a 10 °F approach costs 15.3 % of the annualised cost against the true optimum of 21.5 °F.
- **How flat is the optimum?** Within 5 % of the optimum TAC, T_1 ranges over

$$281.25 \leq T_1 \leq 297.75 \text{ °F} \quad \iff \quad 14.25 \leq \text{approach} \leq 30.75 \text{ °F}$$

- **The practical conclusion.** The optimum is broad — a ± 8 °F window either side costs less than 5 %. So the 10 °F heuristic is defensible as a *starting* design point, and the 15.3 % is the known price of stopping there

rather than optimising. Note also that the approach at the *cooler* hot end is 168.5°F: the two ends of this network are nowhere near equally tight, and a single approach-temperature rule cannot capture that.

6.2 P2 — Is the boiling-point rule really Fenske's equation?

EDM.pdf, slide 96. Toluene/benzene, $T_{avg} = 368.515$ K.

$$\begin{array}{ll} S_{99\%} = 9801, & \ln S = 9.19024 \\ S_{99.9\%} = 998\,001, & \ln S = 13.8135 \end{array}$$

- The value of λ/R that makes the two expressions agree:

$$\frac{\lambda}{R} = 5080.11 \text{ K} \quad \Longleftrightarrow \quad \lambda = 42.236 \text{ kJ/mol} = 10.0942 \text{ kcal/mol}$$

- At that value the two methods agree to three significant figures:

λ/R (K)	$\ln \alpha$	α	Fenske N
5080	1.14403	3.13938	8.033
3800	0.855748	2.35313	10.739
back-of-envelope	—	—	8.047

$$\text{back-of-envelope } N = 8.047 \quad \text{vs.} \quad \text{Fenske } N = 8.033 \quad \Rightarrow \quad \boxed{\text{the claim holds at } \lambda/R \approx 5080 \text{ K}}$$

- **The rule is not unit-dependent.** Repeating the back-of-envelope calculation in **degrees Rankine** gives $N = 8.04706$ — identical. The factor of 3 in the denominator absorbs the scale, which is why the rule survives a change of temperature unit.
- **But it is sensitive to λ .** At $\lambda/R = 3800$ K, Fenske gives 10.74 trays — 33% more than at 5080 K. The rule works because $\lambda/R \approx 5080$ K is typical for the organic separations it is used on, not because it is exact.
- **Reading it backwards is also useful.** With $\lambda/R = 5000$ K ($\lambda = 10$ kcal/mol), the implied separation is $\ln S = 9.06086$, $S = 8611.55$, i.e. a 98.93% **split of each component** — slightly short of the 99% the rule is normally quoted for.

Formula Sheet

EP = product value – raw-material cost
gross profit = value of products – cost of reactants

$$m = \frac{\gamma_s P_s^\circ}{P}$$

$$L = 1.4 mG$$

$$N + 1 = \frac{\ln \left[\left(\frac{y_{in} - m x_{in}}{y_{out} - m x_{in}} \right) \left(1 - \frac{mG}{L} \right) + \frac{mG}{L} \right]}{\ln(L/mG)}$$

$$\text{TAC} = C_s G y_{in} \left(\frac{y_{out}}{y_{in}} \right) 8150 + C_N [N + 2]$$

$$N_{min} \approx \frac{T_{b1} + T_{b2}}{3(T_{b2} - T_{b1})}$$

$$N_{min} = \frac{\log \left[\frac{x_D}{1-x_D} \cdot \frac{1-x_B}{x_B} \right]}{\log \alpha}$$

$$Q = U A \Delta T_{lm},$$

economic potential

per lb of product

equilibrium slope

isothermal dilute absorber (H3)

Kremser

absorber cost model

back-of-envelope trays

Fenske

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} \quad \text{exchanger}$$

The heuristics, in one place

Solute < 5 mol% in the gas	adsorption is the cheapest recovery route
A raw material can be the solvent	feed the process through the absorber
Valuable material	recover > 99%; first guess 99.5%
Isothermal dilute absorber	$L = 1.4mG$, i.e. $1 < L/mG < 2$
Adiabatic absorber	avoid; $L \approx 14mG$
Approach temperature	10 °F from ambient to the organic boiling point
Raw-material share of product cost	normally 33–85%

The order things must be decided in

- 1 Batch or continuous
- 2 Input–output structure (dominates economics; discard any path whose products are worth less than its feeds)
- 3 Recycle structure
- 4a Vapour recovery system — *before* the LSS, since its losses fix the purge composition
- 4b Liquid separation system
- 5 Energy integration — *last*, since HENS needs every stream’s flow, composition and temperature

Prepared from the five CHE F314 slide decks named in the box on page 1; every number is reproduced by `pdp-answers-check.py`, which exits 0. Structure and depth follow the course standard set by `SP2-notes.tex` (CHE F313). `EIA.Selected.Slides.pdf` has no text layer, so section 5 carries only what is legible on its six image pages — that limit is stated there rather than filled in from another source. Symbols: L solvent molar flow, G gas molar flow, m equilibrium slope $y^* = mx$, N theoretical stages, x and y liquid and gas solute mole fractions, γ_s solvent activity coefficient, P_s° solute vapour pressure, EP economic potential, TAC total annualised cost, U overall heat-transfer coefficient, A exchanger area, ΔT_{lm} log-mean temperature difference, ΔT_{min} minimum approach temperature, Q exchanger duty, α relative volatility, N_{min} minimum number of trays, λ latent heat of vaporisation, F stream mass flow, c_p heat capacity, C_A area cost, C_N stage cost, C_s solute cost, VRS vapour recovery system, LSS liquid separation system, EIA energy integration analysis, HENS heat-exchanger network synthesis, BFD block flow diagram, PFD process flow diagram, P&ID piping and instrumentation diagram, HDA hydrodealkylation, DCE dichloroethane, VC vinyl chloride, MIBK methyl isobutyl ketone.